EXP.NO:01

PROBLEM 1 : FIND THE TIME COMPLEXITY USING COUNTER METHOD

PROBLEM STATEMENT:

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

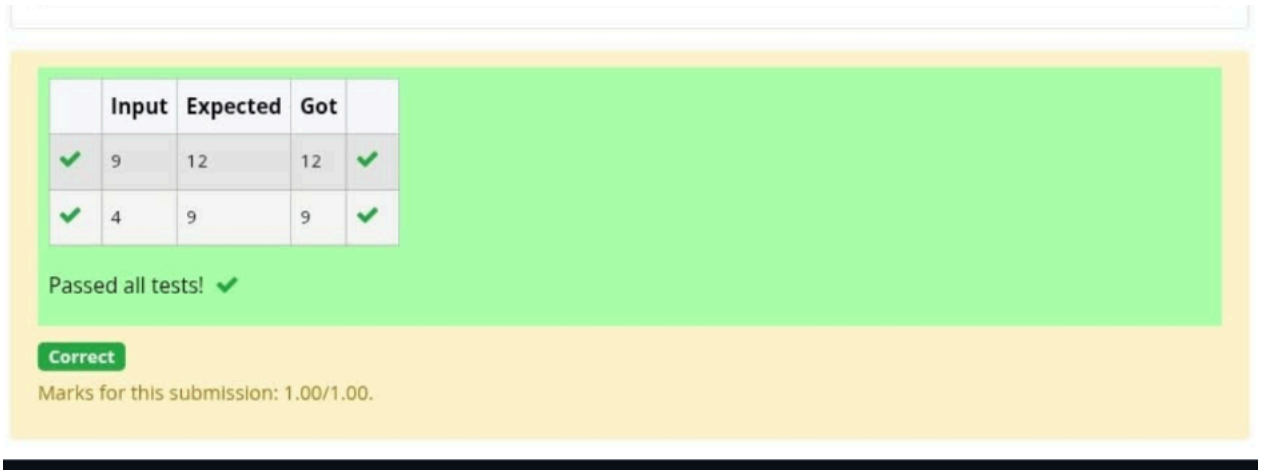
CODE:

```c
#include<stdio.h>

int main(){
    int n;
    scanf("%d", &n);
    int i=1, s=1;
    int count=0;
    while(s<=n){
        count++;
        i++;
        count++;
        s += i;
        count++;
    }
    count+=3;
    printf("%d", count);
    return 0;
}
```

EXP.NO:01

OUTPUT:



	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.